INTRODUCTION

Matter may exist in three different physical states namely solid, liquid and gas. If you look around, you may find mostly solids rather than liquids and gases. Solids differ from liquids and gases by possessing definite volume and definite shape. In the solids the atoms or molecules or ion are tightly held in an ordered arrangement and there are many types of solids such as diamond, metals, plastics etc., and most of the substances that we use in our daily life are in the solid state. We require solids with different properties for various applications. Understanding the relation between the structure of solids and their properties is very much useful in synthesizing new solid materials with different properties.

In this chapter, we study the characteristics of solids, classification, structure and their properties; we also discuss the crystal defects and their significance.

6.1 General characteristics of solids

We have already learnt in XI STD that gas molecules move randomly without exerting reasonable forces on one another. Unlike gases, in solids the atoms , ions or molecules are held together by strong force of attraction. The general characteristics of solids are as follows,

(i) Solids have definite volume and shape.

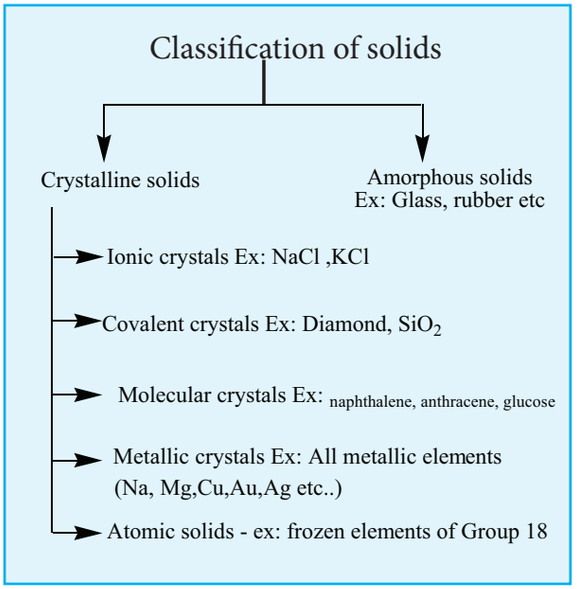
(ii) Solids are rigid and incompressible

(iii) Solids have strong cohesive forces.

(iv) Solids have short inter atomic, ionic or molecular distances.

(v) Their constituents (atoms , ions or molecules) have fixed positions and can only oscillate about their mean positions

6.2 Classification of solids:



We can classify solids into the following two major types based on the arrangement of their constituents.

(i) Crystalline solids

(ii) Amorphous solids.

The term crystal comes from the Greek word “krystallos” which means clear ice. This term was first applied to the transparent quartz stones, and then the name is used for solids bounded by many flat, symmetrically arranged faces.



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